Residual conductivity at isotropic conical band crossings

Abstract

We consider spinless, noninteracting electrons governed by a real periodic Schrödinger operator in two dimensions. Suppose that the Fermi energy meets the spectrum at finitely many isolated, isotropic conical crossings of two adjacent bands. For a constant electric field switched on at time zero, we prove that the Cesàro mean of the linear-response conductivity is $1/16$ times the identity per crossing, in atomic units. The proof starts from the full Schrödinger evolution. An integrated band-energy identity cancels the term linear in time, time reversal cancels the antisymmetric response, and a local two-dimensional spectral reduction determines the remaining contribution. The conical eigenvalue assumptions alone imply the required current matrix elements. We also justify the Stark dynamics, the trace per unit area, and differentiation of the unbounded current for potentials in H^1_{per}.

1 The model and the theorem

At an isolated conical crossing, a two-dimensional semimetal has no Fermi curve, but its response to a weak electric field can have a nonzero long-time average. We prove this statement for a periodic Schrödinger operator under explicit assumptions on its bands. The order of limits is part of the result: we first take the derivative at zero field at each fixed time, and then average that derivative over an increasing time interval.

Let $\mathcal{R} \subset \mathbb{R}^2$ be a lattice, let Ω be a fundamental cell, and define the reciprocal lattice by

$$\mathcal{R}^* = \{K \in \mathbb{R}^2 : K \cdot R \in 2\pi\mathbb{Z} \text{ for all } R \in \mathcal{R}\}.$$

We write $\mathcal{B}$ for a reciprocal fundamental cell and $\mathbb{T}^2_* = \mathbb{R}^2/\mathcal{R}^*$ for the Brillouin torus. In particular,

$$|\Omega|\,|\mathcal{B}| = (2\pi)^2. \tag{1.1}$$

All scalar products are linear in the second argument. Electrons are spinless, and we use atomic units with the signs of the field and current specified below.

Let $V \in H^1_{\mathrm{per}}(\Omega;\mathbb{R})$ and put $p = -i\nabla$. The Hamiltonian and its periodic Bloch fibers are

$$H = \tfrac{1}{2}p^2 + V \quad \text{on } L^2(\mathbb{R}^2), \qquad H_k = \tfrac{1}{2}(p+k)^2 + V \quad \text{on } \mathscr{H} := L^2_{\mathrm{per}}(\Omega). \tag{1.2}$$

Their domains are $H^2(\mathbb{R}^2)$ and $\mathscr{D} := H^2_{\mathrm{per}}(\Omega)$, respectively. Each fiber has compact resolvent. Denote its ordered eigenvalues, counted with multiplicity, by $\lambda_1(k) \le \lambda_2(k) \le \cdots$, and set $\lambda_0 = -\infty$.

Fix a Fermi energy $\mu_{\mathrm{F}} \in \mathbb{R}$. We assume that there are an integer $N \ge 1$ and distinct points $k_1, \dots, k_M \in \mathbb{T}^2_*$ such that

$$\lambda_N(k) < \mu_{\mathrm{F}} < \lambda_{N+1}(k) \qquad \text{for } k \in \mathbb{T}^2_* \setminus \{k_1, \dots, k_M\}, \tag{1.3}$$

whereas at each crossing,

$$\lambda_{N-1}(k_i) < \lambda_N(k_i) = \mu_{\mathrm{F}} = \lambda_{N+1}(k_i) < \lambda_{N+2}(k_i). \tag{1.4}$$

Finally, for some $v_i > 0$,

$$\begin{aligned} \lambda_N(k_i + q) &= \mu_{\mathrm{F}} - v_i|q| + O(|q|^2), \\ \lambda_{N+1}(k_i + q) &= \mu_{\mathrm{F}} + v_i|q| + O(|q|^2) \qquad (q \longrightarrow 0). \end{aligned} \tag{1.5}$$

These are local assertions in Euclidean coordinates on $\mathbb{T}^2_*$. The remainder bounds are uniform in the direction of q. No assumption is made about crossings among occupied bands or among unoccupied bands.

Let $e_\beta, e_\alpha \in \mathbb{R}^2$ specify the field and current directions; they need not be unit vectors. For $\varepsilon \geq 0$, define

$$\begin{aligned} \gamma_0 = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H), \qquad H_\varepsilon = H - \varepsilon x \cdot e_\beta, \qquad J_\alpha = p \cdot e_\alpha, \\ \gamma_\varepsilon(t) = e^{-itH_\varepsilon}\gamma_0 e^{itH_\varepsilon}, \qquad j_\varepsilon(t) = \underline{\mathrm{Tr}}(J_\alpha \gamma_\varepsilon(t)). \end{aligned} \tag{1.6}$$

Here H_ε is the self-adjoint closure of its restriction to $C_c^\infty(\mathbb{R}^2)$, and the trace per unit area of a periodic, locally trace-class operator is

$$\underline{\mathrm{Tr}}(A) = \frac{1}{|\Omega|}\,\mathrm{Tr}(\mathbf{1}_\Omega A \mathbf{1}_\Omega). \tag{1.7}$$

The existence of this closure and of the current trace in (1.6) is proved in Appendix A.

Theorem 1.1. *Under* (1.3)–(1.5), *the current* $j_\varepsilon(t)$ *is well defined for every* $\varepsilon, t \geq 0$, *and* $j_0(t) = 0$. *For every fixed* $t \geq 0$, *the limit*

$$\sigma_{\alpha\beta}(t) := \lim_{\varepsilon \downarrow 0} \frac{j_\varepsilon(t)}{\varepsilon}$$

exists. Moreover,

$$\lim_{T \longrightarrow \infty} \frac{1}{T} \int_0^T \sigma_{\alpha\beta}(t)\, dt = \frac{M}{16}\, e_\alpha \cdot e_\beta. \tag{1.8}$$

Each distinct crossing on the reciprocal torus is counted once. The motivating case is the two-crossing band geometry of spinless graphene; it gives $1/8$ times the identity. The statement concerns the iterated limit in (1.8); it does not require pointwise convergence of $\sigma_{\alpha\beta}(t)$ as $t \to \infty$.

The proof has three parts. We first derive the fixed-time response formula for the full Hamiltonian. We then use the total energy of the occupied bands to cancel the apparent linear growth in time. Finally, we reduce the contribution of each crossing to an elementary integral. The standard operator results used in the argument are drawn from Reed–Simon [1, 2, 3]; the analytic details needed here are included in the appendix.

2 Bloch dynamics and linear response

We use the Bloch transform

$$(\mathcal{U}f)(k, x) = \sum_{R \in \mathcal{R}} e^{-ik\cdot(x+R)} f(x + R), \qquad x \in \Omega, \tag{2.1}$$

with direct-integral measure $dk/|\mathcal{B}|$ and the ordinary $L^2(\Omega, dx)$ norm on each fiber. Lattice Parseval gives its unitarity. In this representation, H and J_α have fibers H_k and $v_\alpha(k)$, respectively; these identities follow on a core and then by closure [3, Theorem XIII.97]. The trace per unit area is

$$\underline{\mathrm{Tr}}(A) = \frac{1}{(2\pi)^2} \int_{\mathcal{B}} \mathrm{Tr}_{\mathscr{H}}\, A(k)\, dk \tag{2.2}$$

for the operators used below. The factor follows from (1.1); a trace-class justification is given in Lemma A.3.

Set

$$P(k) = \mathbf{1}_{(-\infty,\mu_{\mathrm{F}}]}(H_k), \qquad v_\eta(k) = (p+k)\cdot e_\eta = \partial_\eta H_k, \quad \eta \in \{\alpha,\beta\}, \tag{2.3}$$

where $\partial_\eta = e_\eta \cdot \nabla_k$. The projection $P(k)$ has rank N almost everywhere. Its rank is $N+1$ at a crossing, but changes on this finite set do not affect any direct integral or current. All band formulas below are understood away from these points.

For $K \in \mathcal{R}^*$, the unitary $\tau_K u(x) = e^{-iK\cdot x}u(x)$ satisfies

$$H_{k+K} = \tau_K H_k \tau_K^*, \qquad v_\eta(k+K) = \tau_K v_\eta(k)\tau_K^*.$$

Consequently, band energies and the traces below are functions on $\mathbb{T}^2_*$.

Choose an orthonormal eigenbasis $u_n(k)$, measurable in k, and define for $n \le N < m$

$$\begin{aligned} \Delta_{mn}(k) &= \lambda_m(k) - \lambda_n(k), \\ z_{nm}(k) &= \langle u_n(k), v_\alpha(k) u_m(k)\rangle \langle u_m(k), v_\beta(k) u_n(k)\rangle. \end{aligned} \tag{2.4}$$

The response and sum-rule formulas below are independent of the choice of basis within a degenerate eigenspace; the absolute-value estimates hold for every such choice. Existence of a measurable eigenbasis is part of the Bloch spectral theory in [3, Theorem XIII.98].

2.1 Bounds for the band sums

The uniform graph estimates in Lemma A.1 and the boundedness of the occupied eigenvalues imply

$$\sup_{k\in\mathcal{B}} \sum_{n=1}^{N} \|v_\eta(k)u_n(k)\|^2 < \infty. \tag{2.5}$$

Throughout, a supremum involving $P(k)$ may be read as an essential supremum. By Cauchy–Schwarz and Parseval,

$$\sum_{n\le N<m} |z_{nm}(k)| \le \sum_{n=1}^{N} \|v_\alpha(k)u_n(k)\|\,\|v_\beta(k)u_n(k)\| \le C. \tag{2.6}$$

In particular, this estimate controls the infinitely many unoccupied bands without an energy cutoff.

Choose disjoint small disks D_i about the crossings on $\mathbb{T}^2_*$. On the complement of these disks, all occupied-to-unoccupied gaps have a common positive lower bound. Inside D_i, the only gap that can become small is

$$\Delta_i(q) := \Delta_{N+1,N}(k_i+q) = 2v_i|q| + O(|q|^2), \qquad c|q| \le \Delta_i(q) \le C|q|. \tag{2.7}$$

Every other occupied-to-unoccupied gap in these disks is bounded below uniformly. These assertions follow from continuity of the bands, compactness of the torus, and (1.4). Hence

$$k \longmapsto \sum_{n\le N<m} \frac{|z_{nm}(k)|}{\Delta_{mn}(k)} \quad \text{belongs to } L^1(\mathcal{B}). \tag{2.8}$$

The possible singularity is $C/|k-k_i|$, which is integrable in two dimensions.

2.2 The fixed-time response

For each fiber, let $U_{\varepsilon,k}(t,s)$ be the unitary propagator solving

$$i\partial_t U_{\varepsilon,k}(t,s) = H_{k+\varepsilon t e_\beta} U_{\varepsilon,k}(t,s), \qquad U_{\varepsilon,k}(s,s) = I. \tag{2.9}$$

The common domain is $\mathscr{D}$. The temporal gauge gives the exact current formula

$$j_\varepsilon(t) = \frac{1}{(2\pi)^2} \int_{\mathcal{B}} \mathrm{Tr}\big[(v_\alpha(k) + \varepsilon t\, e_\alpha \cdot e_\beta) U_{\varepsilon,k}(t,0) P(k) U_{\varepsilon,k}(t,0)^*\big]\, dk. \tag{2.10}$$

In particular the accelerated momentum is $p + k + \varepsilon t e_\beta$. Both the sign and the scalar term in (2.10) are consequences of the scalar potential $-\varepsilon x \cdot e_\beta$. Lemmas A.2 and A.3 prove the formula for the regularity of V assumed here.

Let $\mathcal{C}$ denote complex conjugation. Since V is real,

$$\mathcal{C} H_k \mathcal{C} = H_{-k}, \qquad \mathcal{C} P(k) \mathcal{C} = P(-k), \qquad \mathcal{C} v_\eta(k) \mathcal{C} = -v_\eta(-k). \tag{2.11}$$

Thus $\mathrm{Tr}(v_\alpha(k)P(k))$ is real and odd under $k \mapsto -k$ on the torus. Formula (2.10) at $\varepsilon = 0$ proves that $j_0(t) = 0$.

Proposition 2.1. *For each fixed $t \geq 0$, the derivative at zero field exists and equals*

$$\begin{aligned} \sigma_{\alpha\beta}(t) = \frac{1}{(2\pi)^2} \int_{\mathcal{B}} \Bigg\{ & N t\, e_\alpha \cdot e_\beta \\ & - 2 \sum_{n \leq N < m} \mathrm{Re}\, z_{nm} \left(\frac{t}{\Delta_{mn}} - \frac{\sin(t\Delta_{mn})}{\Delta_{mn}^2} \right) \\ & + 2 \sum_{n \leq N < m} \mathrm{Im}\, z_{nm} \frac{1 - \cos(t\Delta_{mn})}{\Delta_{mn}^2} \Bigg\}\, dk. \end{aligned} \tag{2.12}$$

All displayed band sums and integrals converge absolutely for fixed t; the same bounds hold uniformly on bounded time intervals.

Proof. The differentiation of (2.10), including the unbounded factor v_α, is justified in Lemma A.4. It gives

$$\sigma_{\alpha\beta}(t) = \frac{1}{(2\pi)^2} \int_{\mathcal{B}} \left\{ N t\, e_\alpha \cdot e_\beta - i \int_0^t s\ \mathrm{Tr}\Big(v_\alpha e^{-i(t-s)H_k} [v_\beta, P] e^{i(t-s)H_k} \Big)\, ds \right\} dk. \tag{2.13}$$

For $r = t - s$, the trace in this formula is

$$\sum_{n \leq N < m} (z_{nm} e^{-ir\Delta_{mn}} - \overline{z_{nm}} e^{ir\Delta_{mn}}). \tag{2.14}$$

Its absolute convergence follows from (2.6). The two elementary integrals

$$\int_0^t s \sin((t-s)\Delta)\, ds = \frac{t}{\Delta} - \frac{\sin(t\Delta)}{\Delta^2}, \qquad \int_0^t s \cos((t-s)\Delta)\, ds = \frac{1 - \cos(t\Delta)}{\Delta^2}$$

give (2.12). Finally,

$$\frac{|\sin(t\Delta)|}{\Delta^2} \leq \frac{t}{\Delta}, \qquad \frac{1 - \cos(t\Delta)}{\Delta^2} \leq \frac{t^2}{2}.$$

Together with (2.6) and (2.8), these inequalities justify the separate terms and all interchanges of sums and integrals. □

3 Cancellation of the secular and antisymmetric terms

The first two lines of (2.12) contain terms proportional to t. Their cancellation is an identity for the total occupied band energy. At the crossings, that energy is not twice differentiable in the classical sense; the following argument accounts for those points.

Lemma 3.1 (Integrated band-energy identity)**.** *One has*

$$N|\mathcal{B}|\, e_\alpha \cdot e_\beta = 2 \int_{\mathcal{B}} \sum_{n \le N < m} \frac{\operatorname{Re} z_{nm}(k)}{\Delta_{mn}(k)}\, dk. \tag{3.1}$$

Proof. On the punctured torus define

$$E(k) = \sum_{n=1}^{N} \lambda_n(k) = \operatorname{Tr}(H_k P(k)).$$

The occupied cluster is isolated there, so its Riesz projection and E are smooth, even at internal band degeneracies. Differentiation of the spectral projection gives

$$\partial_\beta E = \operatorname{Tr}(v_\beta P), \qquad \partial_\alpha \partial_\beta E = N e_\alpha \cdot e_\beta - 2 \sum_{n \le N < m} \frac{\operatorname{Re} z_{nm}}{\Delta_{mn}}. \tag{3.2}$$

For completeness, $P(\partial_\beta P)P = 0$ and $(I-P)(\partial_\beta P)(I-P) = 0$. Differentiating $[H_k, P(k)] = 0$ therefore yields, for $n \le N < m$,

$$\langle u_m, (\partial_\beta P) u_n \rangle = -\frac{\langle u_m, v_\beta u_n \rangle}{\Delta_{mn}}.$$

This and its adjoint give the second identity in (3.2), since $\partial_\beta v_\alpha = e_\alpha \cdot e_\beta$.

The first identity and (2.5) show that $\nabla_k E$ is bounded on the punctured torus, uniformly up to the punctures. The second identity and (2.8) show that its classical Hessian is integrable. Delete disks of radius ρ around the crossings and integrate $\partial_\alpha(\partial_\beta E)$ over the remaining torus. Periodicity cancels the outer cell faces. Each small circle contributes at most $C\rho$, because $\partial_\beta E$ is bounded. Letting $\rho \downarrow 0$ gives

$$\int_{\mathcal{B}} \partial_\alpha \partial_\beta E\, dk = 0.$$

Substitution of (3.2) proves (3.1). In particular, no point-supported correction is left at a conical crossing. □

Time reversal removes the last line of (2.12). Indeed, at $-k$ we may use the eigenvectors $\mathcal{C}u_n(k)$, so that

$$\Delta_{mn}(-k) = \Delta_{mn}(k), \qquad z_{nm}(-k) = \overline{z_{nm}(k)}. \tag{3.3}$$

These identities are understood after summation over any degenerate eigenspaces. Reciprocal-lattice covariance makes the integrands well defined on $\mathbb{T}^2_*$, whether or not the chosen cell $\mathcal{B}$ is symmetric. The imaginary part in (2.12) is integrable and odd, and hence its integral vanishes. The individual matrix elements need not be real.

Combining this observation with Lemma 3.1 yields the response formula used in the rest of the proof:

$$\boxed{\sigma_{\alpha\beta}(t) = \frac{2}{(2\pi)^2} \int_{\mathcal{B}} \sum_{n \le N < m} \operatorname{Re} z_{nm}(k)\, \frac{\sin(t\Delta_{mn}(k))}{\Delta_{mn}(k)^2}\, dk.} \tag{3.4}$$

The estimates already proved imply that this is a continuous function of t on every bounded time interval.

4 The current matrix elements at a conical crossing

The hypotheses describe band energies, while (3.4) depends on eigenvectors through z_{nm}. We now derive the required matrix elements from the energy assumptions.

Lemma 4.1 (Local conical matrix element)**.** *Fix a crossing k_i, write $v = v_i$, and let $\omega = q/|q|$. Then, uniformly in $\omega \in S^1$,*

$$\operatorname{Re} z_{N,N+1}(k_i + q) = v^2 \left(e_\alpha \cdot e_\beta - (e_\alpha \cdot \omega)(e_\beta \cdot \omega)\right) + O(|q|). \tag{4.1}$$

Proof. The two bands meeting at k_i form an isolated spectral cluster in a neighborhood of $q = 0$. Let $Q(q)$ be its rank-two Riesz projection. This projection is real analytic into $\mathcal{B}(\mathscr{H}, \mathscr{D})$. Indeed, for z on a fixed contour enclosing the pair, put $R_0(z) = (H_{k_i} - z)^{-1}$. Then

$$(H_{k_i+q} - z)^{-1} = R_0(z)\big[I + (H_{k_i+q} - H_{k_i})R_0(z)\big]^{-1}.$$

The inverse on the right is a uniformly convergent Neumann series for small complex q, since $H_{k_i+q} - H_{k_i}$ is polynomial from $\mathscr{D}$ to $\mathscr{H}$. Contour integration proves the claim. This is the isolated-cluster construction of analytic perturbation theory [3, Section XII.2]; it does not assert analyticity of the individual conical eigenvalues.

Choose an isometry $W_0 : \mathbb{C}^2 \to \operatorname{Ran} Q(0)$. For small real q, the map

$$W(q) = Q(q)W_0\big(W_0^* Q(q) W_0\big)^{-1/2} \tag{4.2}$$

is a real analytic orthonormal frame for $\operatorname{Ran} Q(q)$, with values in $\mathscr{D}$. Define the Hermitian matrix

$$h(q) = W(q)^* H_{k_i+q} W(q) = a(q)I + d(q) \cdot \boldsymbol{\sigma}, \tag{4.3}$$

where $a(q) \in \mathbb{R}$, $d(q) \in \mathbb{R}^3$, and $\boldsymbol{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$ are the Pauli matrices. Explicitly,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenvalues of h are $a \pm |d|$. Thus (1.5) gives

$$a(q) = \mu_{\mathrm{F}} + O(|q|^2), \qquad d(q) = Lq + O(|q|^2), \qquad L^*L = v^2 I_{\mathbb{R}^2}, \tag{4.4}$$

where $L : \mathbb{R}^2 \to \mathbb{R}^3$ is linear. To see the last assertion, expand the analytic function d at zero and compare $|d(r\omega)| = vr + O(r^2)$ along each ray. It follows that $|L\omega| = v$ for every unit ω, which is precisely the asserted identity. Derivative estimates for the remainders in (4.4) follow from analyticity of a and d.

The physical velocities compressed to this frame are

$$B_\eta(q) = W(q)^* v_\eta(k_i + q) W(q).$$

They must be distinguished from the derivatives of h, because the frame depends on q. With $A_\eta = W^* \partial_\eta W$, differentiation of $H_{k_i+q}W = Wh$ and $W^*W = I$ gives

$$\partial_\eta h = B_\eta + [h, A_\eta]. \tag{4.5}$$

The matrix A_η is bounded, and $h(0) = \mu_{\mathrm{F}} I$. Therefore

$$B_\eta(q) = (Le_\eta) \cdot \boldsymbol{\sigma} + O(|q|). \tag{4.6}$$

For $q \neq 0$, set $n(q) = d(q)/|d(q)|$ and $\pi_\pm(q) = (I \pm n(q)\cdot\boldsymbol{\sigma})/2$. The active bands are simple near the crossing, so

$$z_{N,N+1}(k_i+q) = \mathrm{tr}_{\mathbb{C}^2}\big(\pi_- B_\alpha \pi_+ B_\beta\big). \tag{4.7}$$

For real $a, b, n \in \mathbb{R}^3$ with $|n| = 1$, Pauli multiplication gives

$$\mathrm{Re}\,\mathrm{tr}\Big[\frac{I - n\cdot\boldsymbol{\sigma}}{2}(a\cdot\boldsymbol{\sigma})\frac{I + n\cdot\boldsymbol{\sigma}}{2}(b\cdot\boldsymbol{\sigma})\Big] = a\cdot b - (a\cdot n)(b\cdot n). \tag{4.8}$$

Since $n(r\omega) = L\omega/v + O(r)$, insertion of (4.6) into (4.7) and use of $L^*L = v^2 I$ prove (4.1). $\square$

The angular factor in (4.1) will be denoted by

$$f_{\alpha\beta}(\omega) = e_\alpha\cdot e_\beta - (e_\alpha\cdot\omega)(e_\beta\cdot\omega). \tag{4.9}$$

With $d\omega$ denoting arc length on S^1,

$$\int_{S^1} f_{\alpha\beta}(\omega)\, d\omega = \pi\, e_\alpha\cdot e_\beta. \tag{4.10}$$

Indeed, $\int_{S^1} \omega\otimes\omega\, d\omega = \pi I_{\mathbb{R}^2}$.

5 The Cesàro limit

Time integration of (3.4) gives

$$\frac{1}{T}\int_0^T \sigma_{\alpha\beta}(t)\, dt = \frac{2}{(2\pi)^2}\int_{\mathcal{B}} \sum_{n\le N<m} \mathrm{Re}\, z_{nm}(k)\, \frac{1-\cos(T\Delta_{mn}(k))}{T\Delta_{mn}(k)^3}\, dk. \tag{5.1}$$

Fubini is justified by (2.8), since

$$\frac{1}{T}\int_0^T \frac{|\sin(t\Delta)|}{\Delta^2}\, dt \le \frac{T}{2\Delta}.$$

Keep the fixed disks D_i from Section 2. All transitions outside those disks, and all transitions inside them except the pair $(N, N+1)$, have a common positive gap. By (2.6), their total contribution to (5.1) is $O(T^{-1})$. It remains to evaluate, for one crossing and a fixed sufficiently small $\delta > 0$,

$$I_i(T) = \frac{2}{(2\pi)^2}\int_{|q|<\delta} b_i(q)\, \frac{1-\cos(T\Delta_i(q))}{T\Delta_i(q)^3}\, dq, \qquad b_i(q) = \mathrm{Re}\, z_{N,N+1}(k_i+q). \tag{5.2}$$

Write $q = r\omega$, make the exact change of variables $s = Tr$, and put $D_T(s,\omega) = T\Delta_i((s/T)\omega)$. Then

$$I_i(T) = \frac{1}{(2\pi)^2}\int_{S^1}\int_0^\infty \mathbf{1}_{\{s<\delta T\}}\, \frac{2b_i((s/T)\omega)\, s}{D_T(s,\omega)^3}\big(1-\cos D_T(s,\omega)\big)\, ds\, d\omega. \tag{5.3}$$

The integrand is taken to be zero when $s \ge \delta T$. On its support, (2.7) gives $cs \le D_T(s,\omega) \le Cs$, and b_i is bounded. Consequently its absolute value is at most

$$C\min\{1, s^{-2}\}, \tag{5.4}$$

an integrable function of s independent of T and ω.

For every fixed $s > 0$ and $\omega \in S^1$,

$$D_T(s,\omega) \longrightarrow 2v_i s, \qquad b_i((s/T)\omega) \longrightarrow v_i^2 f_{\alpha\beta}(\omega).$$

Dominated convergence in (5.3) therefore yields

$$\lim_{T\to\infty} I_i(T) = \frac{1}{(2\pi)^2} \int_{S^1} \frac{f_{\alpha\beta}(\omega)}{4v_i}\, d\omega \int_0^\infty \frac{1-\cos(2v_i s)}{s^2}\, ds. \tag{5.5}$$

The scalar integral is πv_i. One self-contained evaluation starts from $u^{-2} = \int_0^\infty y e^{-yu}\, dy$: by Tonelli,

$$\int_0^\infty \frac{1-\cos u}{u^2}\, du = \int_0^\infty y \left(\frac{1}{y} - \frac{y}{1+y^2}\right) dy = \frac{\pi}{2}. \tag{5.6}$$

Using (4.10) in (5.5), we obtain

$$\lim_{T\to\infty} I_i(T) = \frac{1}{(2\pi)^2}\, \frac{\pi}{4}\, \pi e_\alpha \cdot e_\beta = \frac{1}{16}\, e_\alpha \cdot e_\beta. \tag{5.7}$$

Summing over the M crossings and adding the vanishing contribution from the other transitions proves Theorem 1.1.

The calculation also explains the independence from v_i: the two current matrix elements supply a factor v_i^2, while the radial gap and its integration supply the inverse factor. Cesàro averaging is what produces the integrable bound (5.4); no assumption on critical points of the other bands is needed.

A Operator and dynamical details

This appendix verifies the analytic facts used to pass from the Stark evolution to the response formula. Constants may depend on a fixed compact set of momenta and on bounded time and field intervals.

A.1 Domains and self-adjointness

Lemma A.1. *The operators in* (1.2) *are self-adjoint on the stated domains. The fibers have compact resolvent, depend analytically on k as operators from $\mathscr{D}$ to $\mathscr{H}$, and satisfy, uniformly on compact sets of k,*

$$C^{-1}\|u\|_{H^2} \le \|H_k u\| + \|u\| \le C\|u\|_{H^2}. \tag{A.1}$$

Their quadratic-form domains are $H^1_{\mathrm{per}}(\Omega)$. For every real ε, the restriction of H_ε to $C_c^\infty(\mathbb{R}^2)$ is essentially self-adjoint.

Proof. The periodic extension of V is uniformly locally L^2. In dimension two, multiplication by such a potential is infinitesimally $-\Delta$-bounded [3, Theorem XIII.96]:

$$\|Vu\|_2 \le \eta\|\Delta u\|_2 + C_\eta\|u\|_2 \qquad (\eta > 0). \tag{A.2}$$

The same bound holds on the periodic cell, by the localized estimate or by Sobolev interpolation. Kato–Rellich [2, Theorem X.12] gives self-adjointness and (A.1); the first-order terms in k are also infinitesimally $-\Delta$-bounded. Compactness of $H^2_{\mathrm{per}}(\Omega) \hookrightarrow L^2_{\mathrm{per}}(\Omega)$ gives compact resolvent. The polynomial expression for H_k proves analyticity on its fixed domain. Localized Sobolev interpolation also gives

$$|\langle u, Vu\rangle| \le \eta\|\nabla u\|_2^2 + C_\eta\|u\|_2^2,$$

on the full space and on the cell. This identifies the form domains and gives a uniform lower bound on compact sets of momenta.

For the Stark operator, apply the Faris–Lavine theorem [2, Theorem X.38] to

$$2H_\varepsilon = -\Delta + U + W, \qquad U(x) = -2\varepsilon x \cdot e_\beta, \qquad W(x) = 2V(x).$$

Choose $c > 0$ and d so that $U \geq -c|x|^2 - d$. The domain $C_c^\infty(\mathbb{R}^2)$ is dense and invariant under multiplication by x_j and differentiation. The comparison operator $-\Delta + U + W + 2c|x|^2$ is essentially self-adjoint on this domain by [2, Theorem X.29]: after adding a constant, $U + 2c|x|^2$ is nonnegative and locally L^2, while W has relative $-\Delta$ bound zero. Moreover, $-a\Delta + W$ is bounded below for any fixed $0 < a < 1$, by the form estimate above. These are the two hypotheses of Theorem X.38, which proves the assertion. In particular, no global $L^p(\mathbb{R}^2)$ integrability of the periodic potential is used. □

The graph estimate and the bounded occupied energies also give

$$\operatorname*{ess\,sup}_{k\in\mathcal{B}} \|P(k)\|_{\mathcal{B}(\mathscr{H},\mathscr{D})} < \infty. \tag{A.3}$$

A.2 The temporal gauge

Lemma A.2. *The propagators in* (2.9) *exist, preserve* $\mathscr{D}$*, and satisfy*

$$\sup_{\substack{k\in\mathcal{B},\, 0\leq s,t\leq t_* \\ |\varepsilon|\leq\varepsilon_*}} \|U_{\varepsilon,k}(t,s)\|_{\mathcal{B}(\mathscr{D})} < \infty \tag{A.4}$$

for every finite t_*, ε_**. Let* $U_\varepsilon(t,s)$ *be their direct integral, transformed back to* $L^2(\mathbb{R}^2)$*. Then*

$$e^{-itH_\varepsilon} = G_\varepsilon(t)U_\varepsilon(t,0), \qquad G_\varepsilon(t) = e^{i\varepsilon t x\cdot e_\beta}. \tag{A.5}$$

Proof. The Hamiltonians $K(t) = H_{k+\varepsilon t e_\beta}$ have common domain $\mathscr{D}$ and are continuously differentiable in $\mathcal{B}(\mathscr{D},\mathscr{H})$. Their graph norms are uniformly equivalent by (A.1), and

$$\partial_t K(t) = \varepsilon v_\beta(k + \varepsilon t e_\beta).$$

Thus $\partial_t K(t)(K(s) - i)^{-1}$ is bounded and norm continuous, uniformly on the compact parameter sets under consideration. The common-domain evolution theorem [2, Theorem X.70], applied to $1 + iK(t)$ and with the scalar damping removed, gives the propagator. For initial data in $\mathscr{D}$, self-adjointness of $K(t)$ implies $\partial_t \|U_{\varepsilon,k}(t,s)f\|^2 = 0$. Density and the backward construction then give unitarity. The graph estimate in the proof of that theorem gives (A.4); its constants are uniform because

$$\|(K(t) - K(s))(K(s) - i)^{-1}\| \leq C|t - s|.$$

The stepwise propagators used in that construction are measurable in k and respect reciprocal-lattice covariance; both properties pass to their strong limit. The corresponding full-space evolution preserves $H^2(\mathbb{R}^2)$ and solves the equation with Hamiltonian $\frac{1}{2}(p + \varepsilon t e_\beta)^2 + V$.

To identify this evolution with the Stark group, take $f \in H^2(\mathbb{R}^2)$ and put $w(t) = G_\varepsilon(t)U_\varepsilon(t,0)f$. For $\phi \in C_c^\infty(\mathbb{R}^2)$, the vector $G_\varepsilon(t)^*\phi$ is differentiable in H^2 and has compact support. Differentiating the scalar product, using $G_\varepsilon(t)^* p G_\varepsilon(t) = p + \varepsilon t e_\beta$, gives

$$\frac{d}{dt}\langle \phi, w(t)\rangle = -i\langle H_\varepsilon \phi, w(t)\rangle. \tag{A.6}$$

The integrated identity extends to every $\phi \in D(H_\varepsilon)$ because C_c^∞ is a core. To see uniqueness, fix $\phi \in D(H_\varepsilon^2)$ and set $\phi_s = e^{-i(s-t)H_\varepsilon}\phi$. The extended weak identity gives $\partial_s\langle\phi_s, w(s)\rangle = 0$, whence

$$\langle\phi, w(t)\rangle = \langle e^{itH_\varepsilon}\phi, f\rangle = \langle\phi, e^{-itH_\varepsilon}f\rangle.$$

Thus $w(t) = e^{-itH_\varepsilon}f$. Extending in the initial data by continuity proves (A.5) on all of L^2. This weak argument requires no position moment of f. □

A.3 The local current trace

Lemma A.3. *For every finite ε, t, the operator $J_\alpha\gamma_\varepsilon(t)$ has a bounded periodic extension and is locally trace class. Its trace per unit area is* (2.10).

Proof. We first record a consequence of the normalization (2.1). If A is a bounded periodic operator with Hilbert–Schmidt fibers and $\int_{\mathcal{B}}\|A(k)\|_{\mathfrak{S}_2}^2\,dk < \infty$, then

$$\|A\mathbf{1}_\Omega\|_{\mathfrak{S}_2}^2 = \|\mathbf{1}_\Omega A\|_{\mathfrak{S}_2}^2 = \frac{1}{|\mathcal{B}|}\int_{\mathcal{B}}\|A(k)\|_{\mathfrak{S}_2}^2\,dk. \tag{A.7}$$

To prove the first equality with the integral, use an orthonormal basis of $L^2(\Omega)$ extended by zero to the full space. Its Bloch transform at k is that basis multiplied by $e^{-ik\cdot x}$, which is again an orthonormal basis of the fiber. Parseval and Tonelli give the assertion. Applying the result to A^* gives the other equality.

Let Q be a periodic orthogonal projection whose fibers have rank N and are uniformly bounded as maps from $\mathscr{H}$ to $H^1_{\mathrm{per}}(\Omega)$. Then $J_\alpha Q$ is bounded, and its fibers have uniformly bounded Hilbert–Schmidt norms. By (A.7), both $\mathbf{1}_\Omega J_\alpha Q$ and $Q\mathbf{1}_\Omega$ are Hilbert–Schmidt. Their product

$$\mathbf{1}_\Omega J_\alpha Q\mathbf{1}_\Omega = (\mathbf{1}_\Omega J_\alpha Q)(Q\mathbf{1}_\Omega)$$

is trace class [1, Theorem VI.22(h)]. Evaluating the trace with the same cell basis gives

$$\frac{1}{|\Omega|}\operatorname{Tr}(\mathbf{1}_\Omega J_\alpha Q\mathbf{1}_\Omega) = \frac{1}{(2\pi)^2}\int_{\mathcal{B}}\operatorname{Tr}(v_\alpha(k)Q(k))\,dk. \tag{A.8}$$

The exchange of the sum and integral is legitimate because the fiber trace norms are integrable. Localization to any bounded set follows by covering it with finitely many lattice cells and using the same Hilbert–Schmidt factorization.

Apply this to $Q_\varepsilon(t) = U_\varepsilon(t,0)\gamma_0 U_\varepsilon(t,0)^*$. Its fibers have rank N and are uniformly bounded from $\mathscr{H}$ to $\mathscr{D}$ by (A.3) and (A.4). Equation (A.5) gives

$$\gamma_\varepsilon(t) = G_\varepsilon(t)Q_\varepsilon(t)G_\varepsilon(t)^*.$$

Conjugation by $G_\varepsilon(t)$ preserves periodicity: it commutes with each lattice translation up to a scalar phase, which cancels in this expression. It also preserves H^2 for fixed t. Since $G_\varepsilon(t)$ commutes with $\mathbf{1}_\Omega$ and

$$G_\varepsilon(t)^* J_\alpha G_\varepsilon(t) = J_\alpha + \varepsilon t\, e_\alpha \cdot e_\beta,$$

unitary conjugation of the localized trace and (A.8) prove (2.10) and the asserted local trace class. □

A.4 Differentiating the unbounded current

Lemma A.4. *For fixed t, one may differentiate* (2.10) *at $\varepsilon = 0$ under the integral. The result is* (2.13).

Proof. Fix k outside the crossings and a unit vector $u \in \operatorname{Ran} P(k)$. Set $\psi_\varepsilon(t) = U_{\varepsilon,k}(t,0)u$, $\psi_0(t) = e^{-itH_k}u$, and $\delta_\varepsilon(t) = \psi_\varepsilon(t) - \psi_0(t)$. Duhamel's formula on the common domain gives

$$\delta_\varepsilon(t) = -i \int_0^t U_{\varepsilon,k}(t,s) \left(\varepsilon s v_\beta(k) + \frac{\varepsilon^2 s^2}{2} |e_\beta|^2 \right) e^{-isH_k} u \, ds. \tag{A.9}$$

Since occupied unit vectors have uniformly bounded H^2 norms,

$$\|\delta_\varepsilon(t)\|_2 \le C_t|\varepsilon|, \qquad \|\delta_\varepsilon(t)\|_{H^2} \le C_t \quad (|\varepsilon| \le 1), \tag{A.10}$$

uniformly in k and u. The L^2 estimate holds for arbitrary H^2 initial data with a factor $\|u\|_{H^2}$ on the right. Density then gives strong convergence of $U_{\varepsilon,k}(t,s)$ to $e^{-i(t-s)H_k}$, uniformly for $s \in [0,t]$. It follows from (A.9) that

$$\frac{\delta_\varepsilon(t)}{\varepsilon} \longrightarrow \eta(t) := -i \int_0^t s \, e^{-i(t-s)H_k} v_\beta(k) e^{-isH_k} u \, ds \quad \text{in } L^2. \tag{A.11}$$

This L^2 convergence suffices for the current. Interpolation in (A.10) gives $\|\delta_\varepsilon(t)\|_{H^1} \le C_t|\varepsilon|^{1/2}$. The exact expansion

$$\langle \psi_\varepsilon, v_\alpha \psi_\varepsilon \rangle - \langle \psi_0, v_\alpha \psi_0 \rangle = 2 \operatorname{Re} \langle \delta_\varepsilon, v_\alpha \psi_0 \rangle + \langle \delta_\varepsilon, v_\alpha \delta_\varepsilon \rangle \tag{A.12}$$

has a last term bounded by $C_t|\varepsilon|^{3/2}$. Thus its derivative is $2 \operatorname{Re}\langle \eta(t), v_\alpha \psi_0(t) \rangle$. The bounds are uniform in k and in occupied unit vectors, so the difference quotients of the fiber current have a uniform majorant after summing the N occupied states. Dominated convergence justifies differentiation of (2.10).

Summing (A.11) over an occupied orthonormal basis gives the density derivative

$$-i \int_0^t s \, e^{-i(t-s)H_k} [v_\beta(k), P(k)] e^{i(t-s)H_k} \, ds.$$

The scalar term in (2.10) contributes $Nt \, e_\alpha \cdot e_\beta$, giving (2.13). The trace in that formula is well defined: $[v_\beta, P]$ is a bounded finite-rank operator with range in H^1, and e^{-irH_k} preserves H^1, the form domain from Lemma A.1. Multiplication on the left by v_α therefore still yields a finite-rank bounded operator. Its spectral expansion is the absolutely convergent series (2.14). □

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